Project Plan – BestDeal Product Standardization

Project scope: Standardization of product quantities and units

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Version: 0.1

Status: Draft (project introduction), based on the validated debriefing

1. Project Introduction

1.1 Project context

BestDeal offers its members a procurement platform for purchasing non-product-related spend items from more than 50 major suppliers, such as Viking, Manutan and Technische Unie, under favorable conditions. Its assortment contains approximately 1.2 million products, ranging from office supplies and food products to technical products. These products are presented in the BestDeal webshop, which is powered by BestDeal's own Product Information Management (PIM) system. The PIM is fed daily by the individual product feeds of the suppliers.

Across these feeds, the EAN (European Article Number) acts as the common denominator. It makes it possible to recognize the same product at different suppliers and to link it to BestDeal's own metadata. The supplier-specific article number (SKU) is not suitable for this, because every supplier uses its own numbering.

For this project, BestDeal has provided live (not mocked) supplier data, around 1 GB in total. The dataset contains:

- Product feeds from Manutan, Rexel, Isero, TechData and Technische Unie (TU).
- An anonymized order history covering several years of historical sales data.

During the first client meeting it was mentioned that one of the supplier datasets is intentionally unusable. Identifying which one, and why, is part of understanding the data. The client also allows us to mock data where this helps understanding or presentation, and encourages focusing on one or two product categories or example products to keep findings tangible.

1.2 Problem description

The quality and structure of the product data differs per supplier and per product group. Even when two suppliers offer the exact same product (same EAN), the quantity, packaging unit and unit of measurement can be represented differently. Examples from the case:

- One supplier lists a drink as 0.33 L while another lists it as 330 ml.
- Supplier A offers printing paper (EAN X) in 5 packs of 500 sheets for €50, while supplier B sells the same paper per pack of 500 sheets but prices it per sheet at €0.015.
- One supplier provides a characteristic as a quantity (for example voltage or power), while another only provides the unit (volt or watt).

Because of these differences, prices of equivalent products cannot be compared reliably in the webshop, which makes the comparison difficult or even misleading for customers. At the same time, the backend must keep respecting the supplier-specific order quantities: a customer ordering printing paper from supplier A has to order in multiples of 5 packs, while supplier B allows ordering a single pack.

Standardizing the assortment by hand is economically unfeasible given the size of approximately 1.2 million products. BestDeal therefore needs a scalable, largely automated approach.

1.3 Project goal

The goal of this project is to investigate and develop an IT-based prototype that standardizes product units and quantities across the different supplier feeds, so that prices of equivalent products can be accurately compared in the BestDeal webshop, while the original supplier-specific ordering requirements remain available in the backend.

As agreed with the client, the deliverable is the model or processing pipeline itself; no frontend is required. The prototype will be developed and demonstrated on one or two selected product categories, with the aim of showing that the approach can scale to the full assortment.

In scope:

- Analysis of the supplier data feeds, including data quality and identifying the unusable dataset.
- Standardization of product units (for example L and ml, kg and g).
- Standardization of product quantities and packaging units, including a comparable price per standard unit.
- Preserving the original supplier ordering requirements alongside the standardized values.

Out of scope:

- Standardizing product descriptions, images or other product content.
- Cross-selling, upselling and seasonal trend analysis (the other assignment options, not selected).
- Building a webshop frontend or integrating the prototype into the live PIM.

1.4 Research question

Main research question:

How can BestDeal standardize units and quantities across different supplier data feeds so that prices for equivalent products can be accurately compared in the webshop?

Sub-questions:

1. How are quantities, units and packaging information represented in the supplier feeds, which inconsistencies occur most often, and which supplier dataset is unusable?
2. Which standard unit and quantity should be used per product category, and how can EAN be used to match equivalent products across suppliers?
3. How do existing e-commerce, procurement and PIM environments approach unit conversion and quantity normalization? (library method of subQ1)
4. Which techniques (rule-based conversion, pattern recognition, machine learning or AI, or a hybrid) are most suitable for standardizing the different types of inconsistencies?
5. How can the accuracy and scalability of the prototype be evaluated?

Sub-questions

Current Process Research

1- How are quantities, units and packaging information represented in the supplier feeds, which inconsistencies occur most often, and which supplier dataset is unusable?
How do existing e-commerce, procurement and PIM environments approach unit conversion and quantity normalization? (library method of subQ1)

2- How does the current PIM (Product Information Management) system handle incoming supplier data?

Requirements Research

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Tooling & Feasibility Research

4. Which techniques (rule-based conversion, pattern recognition, machine learning or AI, or a hybrid) are most suitable for standardizing the different types of inconsistencies?

Validation Research

5. How can the accuracy and scalability of the prototype be evaluated?

6.     

Sub-question 1

How are quantities, units and packaging information represented in the supplier feeds, which inconsistencies occur most often, and which supplier dataset is unusable?

Strategy	**Method**	**Why**	**Deliverable**
Field 	Interview (with client)	To understand real workflow behavior	Interview summary
Library 	How does existing PIM (Product Information Management) work? (Available product analysis)	To have an idea on how the solution will be integrated	Research summary (with references) (taken by **Sajad**)
Lab 	Investigating the dataset & finding the unusable dataset (Observation)	To understand the dataset and to see determine feasibility	Observation notes ***I want this one right here - goktan***

Sub-question 2

How does the current PIM (Product Information Management) system handle incoming supplier data?

Strategy	**Method**	**Why**	**Deliverable**
Field 	2nd Interview with the client	Showcasing the design sketch that includes our solution (all visualized)	Interview summary
Showroom 	Peer review on the other two research methods to share the work process	To make sure the team and the client are on the same page and updated about the project progress	Peer review notes

Workshop 	Design a sketch of current PIM workflow	To visualize the current process so that we understand it better	Design diagram (taken by **Miran**)

Source

C2Value - Project Case (2026 Spring), BestDeal case description and assignment; BestDeal debriefing v1.0; client meeting notes and first meeting notes.